AAE 43900 Rocket Propulsion
3 credits

1 Course Information/Description

Flight mechanics and the propulsion system requirements for terrestrial and interplanetary flight. Converging-diverging nozzle analysis and design. Thermochemistry of propellants and performance calculations. Solid rocket motor interior ballistics and design. Thrust chamber and turbo machinery analysis and design. Liquid rocket engine cycle analysis and design. Elements of nuclear and electric propulsion systems.

2 Learning Resources and Reference Text

Brightspace Learning Management System:

Access the course videos, materials, and resources via Purdue's BS learning management system. Begin with the Start Here tab, which describes how the course Brightspace is organized. We strongly suggest that you explore and become familiar not only with the site navigation but with content and resources available for this course. See the Student Services widget on the campus homepage for resources such as Technology Help, Academic Help, Campus Resources, and Protect Purdue.

Textbook:

Heister S. D., Anderson W. E., Pourpoint T. L., Cassady R. J., Rocket Propulsion, Cambridge University Press - Online ISBN: 9781108381376.

Also use the AAE Propulsion Website: https://engineering.purdue.edu/~propulsi/propulsion/

3 Assignments

- All homework assignments will be due on Gradescope by the date and time noted on the assignments.
- A tutorial on how to submit assignments to Gradescope can be found here: https://www.purdue.edu/innovativelearning/supporting-instruction/instructional-technology/docs/Vendor_Guide_for_Student_Submissions.pdf
- Late assignments will be not be accepted for any reason although the lowest homework score will be dropped.
- Homework must be neat, legible, and organized. You must understand the work you turn in.
- In general, we will not share solutions of the homework assignments, instead, you should join our office hours to ask for clarifications.
- You can expect nine homework assignments (each requiring about 1 week to complete) this semester with the **nominal** dates listed on our Brightspace page (see below). These nominal dates are subject to change but given to you to help you plan your work.
- Copying someone else's work is not allowed - copied homeworks will receive a grade of zero and a report will be submitted to the Office of the Dean of Students.

4 Course Learning Objectives

1. Determine velocity gain required to accomplish various rocket vehicle missions
2. Develop capability to predict rocket performance (specific impulse and thrust) from first principles
3. Develop preliminary designs of rocket vehicles using the rocket equation to estimate propellant requirements
4. Develop working understanding of solid, liquid, and hybrid chemical propulsion systems
5. Develop working understanding of rocket turbomachinery and/or electric propulsion fundamentals depending on time/interest

5 Evaluation

- Homework – 20%
- Launch project – 10% (25% of this grade will come from peer evaluations from other members on the launch project team)
- Midterm Exam – 35%
- Final exam – 35%

A grade is assigned at the end of the semester based on a curved scale of all students in class. Those who have demonstrated an excellent understanding of rocket propulsion will be given an A+, A, or A- grade. Those who have demonstrated a minimum acceptable understanding will receive a C+, C, or C- grade with B grades falling in between these extremes. Those who have demonstrated an unacceptable level of understanding will receive a D or F grade.